

Generative Adversarial Nets

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In the original paper[2], the authors presented a novel framework for estimating generative models (models that can generate new data like images, text, etc.) through an adversarial process, where two models are trained simultaneously. Here, adversarial process refers to a situation in which two models or agents are competing against each other with opposing goals. In the context of Generative Adversarial Networks (GANs), this means that the Generative Model (G) will create fake data that looks as real as possible while the Discriminative Model (D) will be distinguishing between real and fake data by estimating the probability of whether a given sample comes from the real training set or generated by G.

In this framework, the training procedure for G is designed to maximize the probability of D making a mistake, which efficiently turns the problem into a minimax two-player game. This makes the ideal outcome of the adversarial process in GAN unique in the space of arbitrary functions G and D, where G will generate data that closely matches the real data distribution (thus, recovers the training data distribution) while D gives a probability of 1/2 for every data sample. Moreover, in GANs, both G and D are neural networks, so they can be trained using backpropagation to update their parameters based on how well they perform, without the need for complex methods like Markov chains or approximate inference networks. This makes GANs more efficient and simpler to train compared to other generative models.

Here, both G and D are multilayer perceptrons and the objective function for this minimax-game training process is

$$\min_G \max_D V(D, G) = E_{x \sim p_{\text{data}}(x)} [\log D(x)] + E_{z \sim p_z(z)} [\log(1 - D(G(z)))] \quad (1)$$

The paper also presents a theoretical analysis of adversarial nets, showing that with sufficient capacity, both G and D will recover the data generating distribution. This process can be implemented using an iterative numerical approach, where the models are updated in alternating steps: k steps of optimizing D and 1 step of optimizing G. This approach ensures that D remains close to its optimal solution, while G improves gradually. It's analogous to Markov Chain Monte Carlo (MCMC) training, where samples are maintained across learning steps to avoid overfitting or burning in the chain.

In practice, Equation 1 might not provide enough gradient early in training when G is weak, as D easily rejects clearly fake data. To address this, G is trained to maximize $\log D(G(z))$, instead of minimizing $\log(1 - D(G(z)))$. This adjustment strengthens the gradients early in training, leading to better learning. After several training steps, if G and D are sufficiently powerful, the discriminator will no longer be able to differentiate between real and fake data, resulting in $p_g = p_{\text{data}}$ and $D(x) = \frac{1}{2}$.

The generator G implicitly defines a probability distribution p_G as the distribution of the samples $G(z)$ obtained when $z \sim p_z$. The goal is for Algorithm 1 to converge to a good estimator of the data distribution, given sufficient capacity and training time. This section assumes a nonparametric setting, where we consider models with infinite capacity. The authors first consider the optimal discriminator D for a fixed generator G.

Proposition 1. For fixed G, the optimal discriminator D^* is:

$$D_G^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_G(x)} \quad (2)$$

Proof. The training criterion for the discriminator D, given any generator G, is to maximize the quantity $V(G, D)$:

$$V(G, D) = \int_x p_{\text{data}}(x) \log(D(x)) dx + \int_z p_z(z) \log(1 - D(G(z))) dz \quad (3)$$

The function $y \rightarrow a \log(y) + b \log(1 - y)$ achieves its maximum at $\frac{a}{a+b}$, which shows that the optimal $D^*(x)$ is the ratio of $p_{\text{data}}(x)$ to $p_{\text{data}}(x) + p_G(x)$.

Theorem 1. The global minimum of the virtual training criterion $C(G)$ is achieved if and only if $p_G = p_{\text{data}}$, and at that point, $C(G) = -\log 4$.

Proof. For $p_G = p_{\text{data}}$, $D^*(x) = \frac{1}{2}$, and thus:

$$C(G) = \log \frac{1}{2} + \log \frac{1}{2} = -\log 4$$

This is the best possible value for $C(G)$, reached only when $p_G = p_{\text{data}}$, proving that the global minimum of $C(G)$ is $-\log 4$.

Proposition 2. If G and D have enough capacity, and at each step of Algorithm 1, the discriminator is allowed to reach its optimum given G, and p_g is updated to improve the criterion:

$$E_{x \sim p_{\text{data}}} [\log D_G^*(x)] + E_{x \sim p_g} [\log(1 - D_G^*(x))]$$

then p_g converges to p_{data} .

Proof. Consider $V(G, D) = U(p_g, D)$ as a function of p_g . Note that $U(p_g, D)$ is convex in p_g . The subderivatives of a supremum of convex functions include the derivative of the function at the point where the maximum is attained. In other words, for $f(x) = \sup_{\alpha \in A} f_{\alpha}(x)$, if $f_{\alpha}(x)$ is convex for every α , then $\partial f(x) \in \partial f$ if $\beta = \arg \sup_{\alpha \in A} f_{\alpha}(x)$. This is equivalent to computing a gradient descent update for p_g at the optimal D given the corresponding G. Since $\sup_D U(p_g, D)$ is convex in p_g with a unique global optimum, and with sufficiently small updates, p_g converges to p_{data} , concluding the proof. In practice, adversarial nets represent a limited family of p_g distributions via the function $G(z; \theta_g)$, and we optimize θ_g rather than p_g itself.

The authors trained adversarial nets on several datasets, including MNIST[5], the Toronto Face Database (TFD)[6], and CIFAR-10[4]. The generator networks used a mix of rectifier linear and sigmoid activations, while the discriminator network used maxout[1] activations and dropout[3] during training. Noise was only applied to the bottommost layer of the generator. To evaluate the performance, we estimate the probability of the test set data under p_g by fitting a Gaussian Parzen window (a non-parametric way to estimate the probability density function (PDF) of a random variable based on observed data) to the samples generated by G, and report the log-likelihood. The results are shown in Table 1, where the log-likelihood estimates for adversarial nets are competitive with other models.

Table 1: Parzen window-based log-likelihood estimates.

Model	MNIST	TFD
DBN [3]	138 ± 2	1909 ± 66
Stacked CAE [3]	121 ± 1.6	2110 ± 50
Deep GSN [6]	214 ± 1.1	1890 ± 29
Adversarial nets	225 ± 2	2057 ± 26

This new framework has both advantages and disadvantages compared to previous generative models. The advantages include: No need for Markov chains, training is simpler since only backpropagation is needed, no inference is required during learning, and a wide variety of functions can be incorporated into the model. On the other hand, the main disadvantages are: The lack of explicit representation for $p_g(x)$, the need to carefully synchronize the discriminator D with the generator G during training, to prevent issues like the "Helvetica scenario," where G collapses too many values of z to the same value of x, reducing diversity.

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